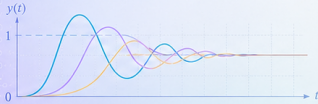
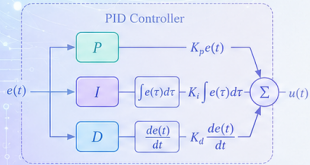
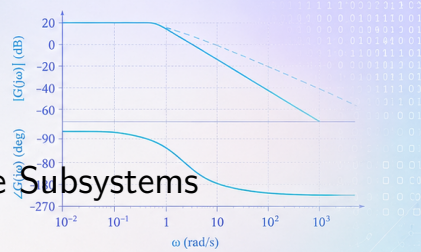


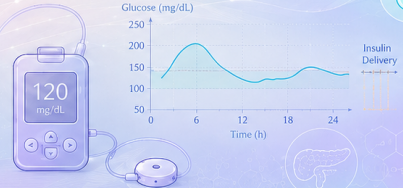
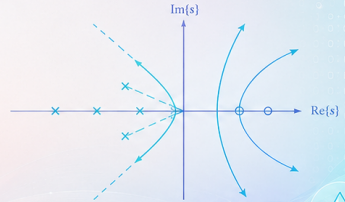
Reduction of Multiple Subsystems

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$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \dots + b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



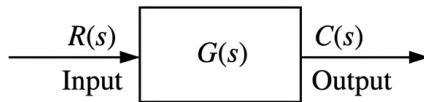
Block Diagrams

- ▶ A subsystem is represented as a block with an input, an output, and a transfer function.
- ▶ Many systems are composed of multiple subsystems.
- ▶ When multiple subsystems are interconnected, a few more schematic elements must be added to the block diagram.

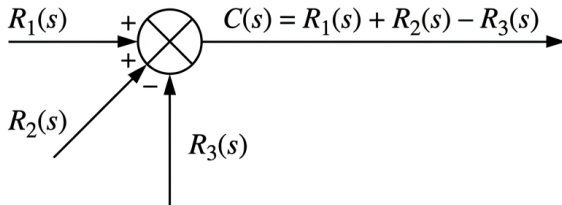
Components of a block diagram for a linear, time-invariant system



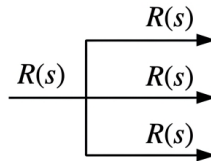
Signals
(a)



System
(b)

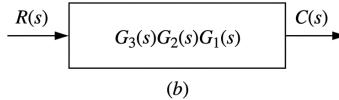
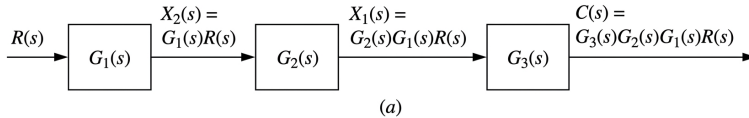


Summing junction
(c)



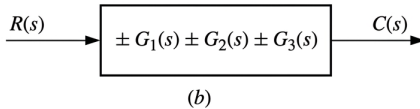
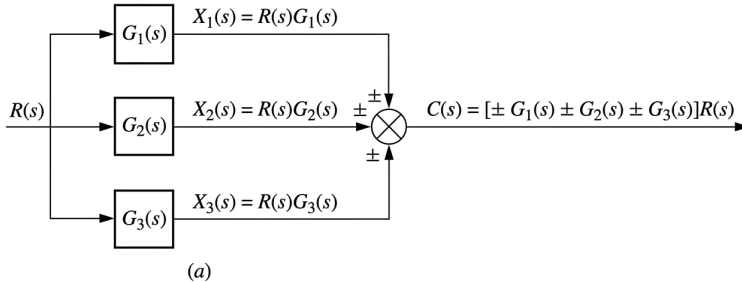
Pickoff point
(d)

Cascade Form



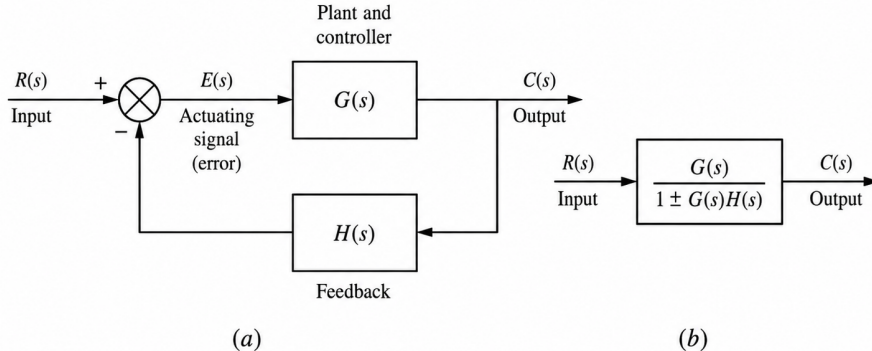
(a) Cascaded subsystems; (b) equivalent transfer function

Parallel Form



(a) Parallel subsystems; (b) equivalent transfer function

Feedback Form



(a) Feedback subsystems; (b) equivalent transfer function

The product, $G(s)H(s)$, is called the **open-loop transfer function, or loop gain**.

Closed-Loop Transfer Function Derivation

The equivalent transfer function is

$$G_e(s) = \frac{C(s)}{R(s)}.$$

For a negative feedback system,

$$E(s) = R(s) - B(s).$$

Since the feedback path contains $H(s)$,

$$B(s) = H(s)C(s).$$

Therefore,

$$E(s) = R(s) - H(s)C(s).$$

The output is

$$C(s) = G(s)E(s).$$

Substituting $E(s)$,

$$C(s) = G(s) [R(s) - H(s)C(s)].$$

Expanding,

$$C(s) = G(s)R(s) - G(s)H(s)C(s).$$

Rearranging,

$$C(s) + G(s)H(s)C(s) = G(s)R(s).$$

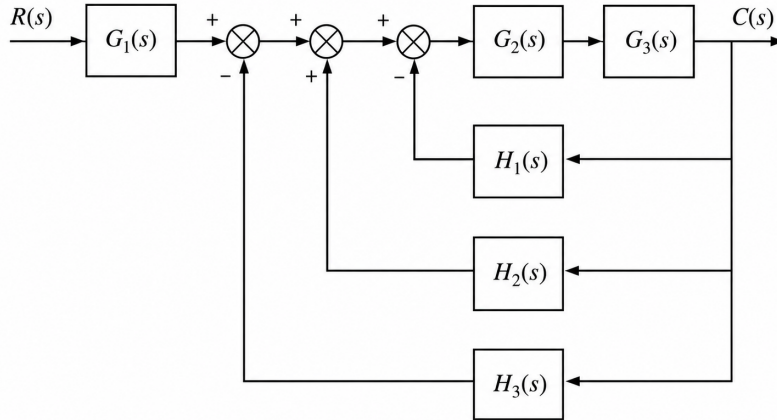
Thus,

$$C(s) [1 + G(s)H(s)] = G(s)R(s).$$

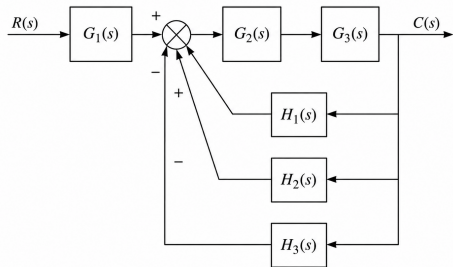
Hence,

$$G_e(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

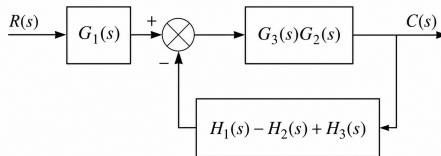
Example: Find equivalent transfer function



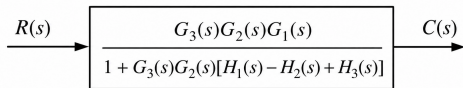
Solution



(a)

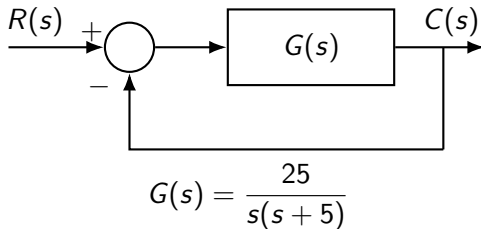


(b)



(c)

Example: Finding Transient Response

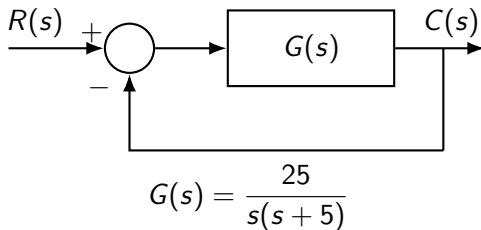


Problem

For the unity-feedback system $G(s)$, determine

- ▶ peak time, T_p ;
- ▶ percent overshoot, PO ;
- ▶ settling time, T_s .

Example: Finding Transient Response



For unity negative feedback,

$$T(s) = \frac{G(s)}{1 + G(s)}.$$

Therefore,

$$\begin{aligned} T(s) &= \frac{\frac{25}{s(s + 5)}}{1 + \frac{25}{s(s + 5)}} \\ &= \boxed{\frac{25}{s^2 + 5s + 25}}. \end{aligned}$$

Problem

For the unity-feedback system $G(s)$, determine

- ▶ peak time, T_p ;
- ▶ percent overshoot, PO ;
- ▶ settling time, T_s .

Compare

$$T(s) = \frac{25}{s^2 + 5s + 25}$$

with the standard second-order transfer function

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

From the constant term,

$$\omega_n^2 = 25,$$

therefore,

$$\boxed{\omega_n = 5 \text{ rad/s}}.$$

From the coefficient of s ,

$$2\zeta\omega_n = 5.$$

Substituting $\omega_n = 5$,

$$2\zeta(5) = 5,$$

Since $0 < \zeta < 1$, the system is **underdamped**.

Peak Time

For an underdamped second-order system,

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}.$$

Substituting

$$\omega_n = 5, \quad \zeta = 0.5,$$

gives

$$T_p = \frac{\pi}{5\sqrt{1 - (0.5)^2}} \approx 0.726 \text{ s}.$$

Peak Time

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gives

$$T_p = \frac{\pi}{5\sqrt{1 - (0.5)^2}} \approx 0.726 \text{ s}.$$

Percent Overshoot

$$PO = 100 \exp\left(-\frac{\zeta\pi}{\sqrt{1 - \zeta^2}}\right).$$

Hence,

$$PO = 100 \exp\left(-\frac{0.5\pi}{\sqrt{1 - (0.5)^2}}\right).$$

Therefore,

$$PO \approx 16.3\%$$

For the 2% settling-time criterion,

$$T_s \approx \frac{4}{\zeta \omega_n}.$$

Substituting

$$\zeta = 0.5, \quad \omega_n = 5,$$

we obtain

$$T_s = \frac{4}{(0.5)(5)} = \frac{4}{2.5}.$$

Therefore,

$$T_s = 1.6 \text{ s}$$

Thank You

Questions and Discussion